

2021Fall AMC 10B**Problem 1**

What is the value of $1234 + 2341 + 3412 + 4123$?

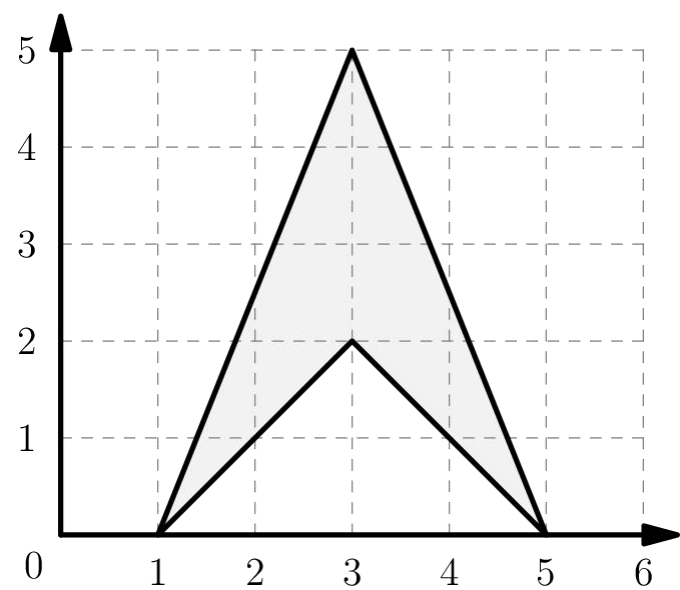
表达式 $1234 + 2341 + 3412 + 4123$ 的值是多少?

- (A) 10,000 (B) 10,010 (C) 10,110 (D) 11,000 (E) 11,110

Problem 2

What is the area of the shaded figure shown below?

下图中阴影部分的面积是多少?



- (A) 4 (B) 6 (C) 8 (D) 10 (E) 12

Problem 3

The expression $\frac{2021}{2020} - \frac{2020}{2021}$ is equal to the fraction $\frac{p}{q}$ in which p and q are positive integers whose greatest common divisor is 1. What is p ?

表达式 $\frac{2021}{2020} - \frac{2020}{2021}$ 等于分数 $\frac{p}{q}$, 其中 p 和 q 是正整数, 它们的最大公约数是 1。问 p 是多少?

- (A) 1 (B) 9 (C) 2020 (D) 2021 (E) 4041

Problem 4

At noon on a certain day, Minneapolis is N degrees warmer than St. Louis. At 4:00 the temperature in Minneapolis has fallen by 5 degrees while the temperature in St. Louis has risen by 3 degrees, at which time the temperatures in the two cities differ by 2 degrees. What is the product of all possible values of N ?

某一天的中午, Minneapolis 的气温比 St. Louis 高 N 度。4:00 时, Minneapolis 的气温下降了 5 度, 而 St. Louis 的气温上升了 3 度, 此时两个城市的气温相差 2 度。问 N 的所有可能值的乘积是多少?

- (A) 10 (B) 30 (C) 60 (D) 100 (E) 120

Problem 5

Let $n = 8^{2022}$. Which of the following is equal to $\frac{n}{4}$?

设 $n = 8^{2022}$ 。问以下哪一项等于 $\frac{n}{4}$?

- (A) 4^{1010} (B) 2^{2022} (C) 8^{2018} (D) 4^{3031} (E) 4^{3032}

Problem 6

The least positive integer with exactly 2021 distinct positive divisors can be written in the form $m \cdot 6^k$, where m and k are integers and 6 is not a divisor of m . What is $m + k$?

恰好有 2021 个不同正约数的最小正整数可以写成 $m \cdot 6^k$ 的形式，其中 m 和 k 是整数，并且 6 不是 m 的约数。问 $m + k$ 的值是多少？

- (A) 47 (B) 58 (C) 59 (D) 88 (E) 90

Problem 7

Call a fraction $\frac{a}{b}$, not necessarily in the simplest form special if a and b are positive integers whose sum is 15. How many distinct integers can be written as the sum of two, not necessarily different, special fractions?

如果 a 和 b 是正整数，总和为 15，那么分数 $\frac{a}{b}$ ，无论是否为最简形式，称为“特殊的”。问有多少个不同的整数可以写成两个特殊分数的和，这里的两个特殊分数不要求互异？

- (A) 9 (B) 10 (C) 11 (D) 12 (E) 13

Problem 8

The largest prime factor of 16384 is 2 because $16384 = 2^{14}$. What is the sum of the digits of the greatest prime number that is a divisor of 16383?

16,384 的约数中的最大素数是 2，因为 $16384 = 2^{14}$ 。问 16,383 的约数中的最大素数的各位数字之和是多少？

- (A) 3 (B) 7 (C) 10 (D) 16 (E) 22

Problem 9

The knights in a certain kingdom come in two colors. $\frac{2}{7}$ of them are red, and the rest are blue.

Furthermore, $\frac{1}{6}$ of the knights are magical, and the fraction of red knights who are magical is 2 times the fraction of blue knights who are magical. What fraction of red knights are magical?

某个王国的骑士有两种颜色： $\frac{2}{7}$ 是红色的，其余的是蓝色的。此外， $\frac{1}{6}$ 的骑士是有魔力的，并且红色骑士中有魔力的比例是蓝色骑士中有魔力的比例的2倍。问红色骑士中有魔力的比例是多少？

- (A) $\frac{2}{9}$ (B) $\frac{3}{13}$ (C) $\frac{7}{27}$ (D) $\frac{2}{7}$ (E) $\frac{1}{3}$

Problem 10

Forty slips of paper numbered 1 to 40 are placed in a hat. Alice and Bob each draw one number from the hat without replacement, keeping their numbers hidden from each other. Alice says, "I can't tell who has the larger number." Then Bob says, "I know who has the larger number." Alice says, "You do? Is your number prime?" Bob replies, "Yes." Alice says, "In that case, if I multiply your number by 100 and add my number, the result is a perfect square." What is the sum of the two numbers drawn from the hat?

帽子里有40张编号为1至40的纸条。Alice和Bob每人从帽子中抽取了一个数，不再放回，并且向对方隐藏。Alice说：“我不知道谁的数更大。”然后Bob说：“我知道谁的数更大。”Alice说：“你知道了？你的数是素数吗？”Bob回答说：“是的。”Alice说：“那么如果我将你的数乘以100加上我的数，结果就是一个完全平方数。”问从帽子里抽取的两个数的和是多少？

- (A) 27 (B) 37 (C) 47 (D) 57 (E) 67

Problem 11

A regular hexagon of side length 1 is inscribed in a circle. Each minor arc of the circle determined by a side of the hexagon is reflected over that side. What is the area of the region bounded by these 6 reflected arcs?

边长 1 的正六边形内接于一个圆。由六边形每条边确定的圆的劣弧都沿着对应的边反射。那么由这 6 条反射后的弧围成的区域的面积是多少？

- (A) $\frac{5\sqrt{3}}{2} - \pi$ (B) $3\sqrt{3} - \pi$ (C) $4\sqrt{3} - \frac{3\pi}{2}$ (D) $\pi - \frac{\sqrt{3}}{2}$ (E) $\frac{\pi + \sqrt{3}}{2}$

Problem 12

Which of the following conditions is sufficient to guarantee that integers x , y , and z satisfy the equation

以下哪个条件可以确保整数 x 、 y 和 z 满足方程

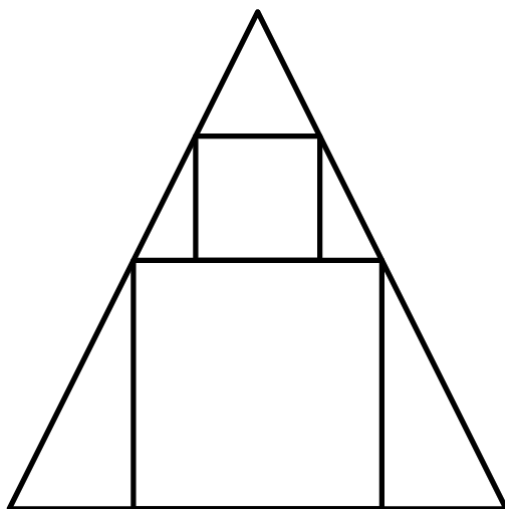
$$x(x - y) + y(y - z) + z(z - x) = 1?$$

- (A) $x > y$ and $y = z$
- (B) $x = y - 1$ and $y = z - 1$
- (C) $x = z + 1$ and $y = x + 1$
- (D) $x = z$ and $y - 1 = x$
- (E) $x + y + z = 1$

Problem 13

A square with side length 3 is inscribed in an isosceles triangle with one side of the square along the base of the triangle. A square with side length 2 has two vertices on the other square and the other two on sides of the triangle, as shown. What is the area of the triangle?

一个边长为 3 的正方形内接于一个等腰三角形，其中正方形的一条边在三角形的底边上。如图所示，一个边长为 2 的正方形的两个顶点在另一个正方形上，而其余两个顶点在三角形的边上。问三角形的面积是多少？



- (A) $19\frac{1}{4}$ (B) $20\frac{1}{4}$ (C) $21\frac{3}{4}$ (D) $22\frac{1}{2}$ (E) $23\frac{3}{4}$

Problem 14

Una rolls 6 standard 6-sided dice simultaneously and calculates the product of the 6 numbers obtained. What is the probability that the product is divisible by 4?

Una 同时掷出 6 个标准的有 6 个面的骰子，并计算得到的 6 个数的乘积。问乘积能被 4 整除的概率是多少？

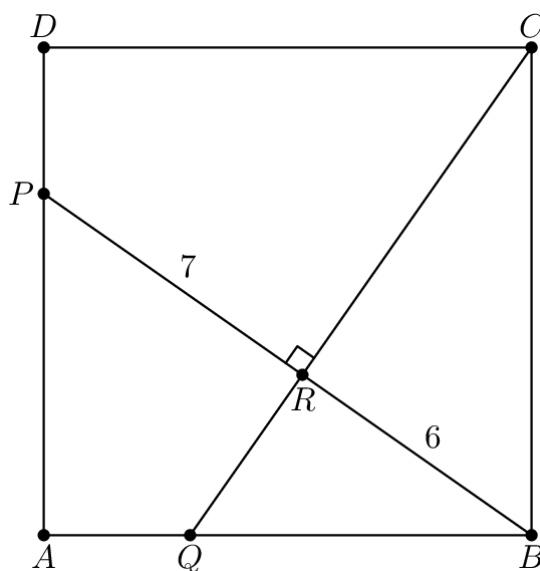
- (A) $\frac{3}{4}$ (B) $\frac{57}{64}$ (C) $\frac{59}{64}$ (D) $\frac{187}{192}$ (E) $\frac{63}{64}$

Problem 15

In square $ABCD$, points P and Q lie on $\overline{AD}$ and $\overline{AB}$, respectively.

Segments $\overline{BP}$ and $\overline{CQ}$ intersect at right angles at R , with $BR = 6$ and $PR = 7$. What is the area of the square?

在正方形 $ABCD$ 中，点 P 和 Q 分别在 $\overline{AD}$ 和 $\overline{AB}$ 上。线段 $\overline{BP}$ 和 $\overline{CQ}$ 在 R 处以直角相交，并且 $BR = 6$ ， $PR = 7$ 。问正方形的面积是多少？



- (A) 85 (B) 93 (C) 100 (D) 117 (E) 125

Problem 16

Five balls are arranged around a circle. Chris chooses two adjacent balls at random and interchanges them. Then Silva does the same, with her choice of adjacent balls to interchange being independent of Chris's. What is the expected number of balls that occupy their original positions after these two successive transpositions?

有五个球沿着圆周摆放。Chris 随机选择两个相邻的球，并互换它们的位置。然后 Silva 进行了同样的操作，并且她对互换的相邻两个球的选择与 Chris 的选择是独立的。问经过这两次换位后，仍然在其原始位置的球的数量的期望值是多少？

- (A) 1.6 (B) 1.8 (C) 2.0 (D) 2.2 (E) 2.4

Problem 17

Distinct lines ℓ and m lie in the xy -plane. They intersect at the origin. Point $P(-1, 4)$ is reflected about line ℓ to point P' , and then P' is reflected about line m to point P'' . The equation of line ℓ is $5x - y = 0$, and the coordinates of P'' are $(4, 1)$. What is the equation of line m ?

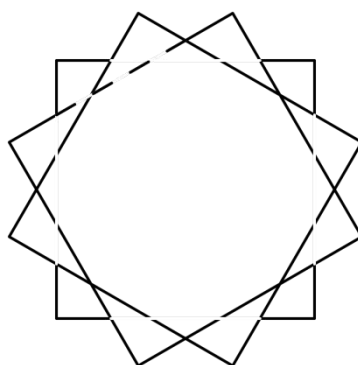
ℓ 和 m 是 xy 坐标平面上的两条不同直线。它们在原点相交。点 $P(-1, 4)$ 沿直线 ℓ 反射到点 P' , 然后 P' 沿直线 m 反射到点 P'' 。直线 ℓ 的方程为 $5x - y = 0$, 而 P'' 的坐标为 $(4, 1)$ 。问直线 m 的方程是什么?

- (A) $5x + 2y = 0$ (B) $3x + 2y = 0$ (C) $x - 3y = 0$
 (D) $2x - 3y = 0$ (E) $5x - 3y = 0$

Problem 18

Three identical square sheets of paper each with side length 6 are stacked on top of each other. The middle sheet is rotated clockwise 30° about its center and the top sheet is rotated clockwise 60° about its center, resulting in the 24-sided polygon shown in the figure below. The area of this polygon can be expressed in the form $a - b\sqrt{c}$, where a , b , and c are positive integers, and c is not divisible by the square of any prime. What is $a + b + c$?

三张相同的边长为 6 的正方形纸片叠放在一起。然后中间的纸片绕其中心顺时针旋转 30° , 最上面的纸片绕其中心顺时针旋转 60° , 形成如下图所示的 24 边形。这个多边形的面积可以用 $a - b\sqrt{c}$ 的形式表示, 其中 a 、 b 和 c 是正整数, 并且 c 不能被任何素数的平方整除。问 $a + b + c$ 是多少?



- (A) 75 (B) 93 (C) 96 (D) 129 (E) 147

Problem 19

Let N be the positive integer $7777 \dots 777$, a 313-digit number where each digit is a 7.

Let $f(r)$ be the leading digit of the r th root of N . What is

$$f(2) + f(3) + f(4) + f(5) + f(6)?$$

设 N 是 313 位的正整数 $7777 \dots 777$, 其中每个数字都是 7。令 $f(r)$ 是 N 的 r 次方根的首位数字。求 $f(2) + f(3) + f(4) + f(5) + f(6)$?

- (A) 8 (B) 9 (C) 11 (D) 22 (E) 29

Problem 20

In a particular game, each of 4 players rolls a standard 6-sided die. The winner is the player who rolls the highest number. If there is a tie for the highest roll, those involved in the tie will roll again and this process will continue until one player wins. Hugo is one of the players in this game. What is the probability that Hugo's first roll was a 5, given that he won the game?

在一个特定的游戏中, 4 个玩家的每个人都抛掷一个标准的 6 个面的骰子。获胜者是掷出最高数的玩家。如果掷出最高数的不止一人, 则这些人将再次抛掷骰子, 并且这一过程将一直持续到有一名玩家获胜为止。Hugo 是这个游戏的一个玩家。假设 Hugo 在游戏中获胜, 那么他第一次掷出的数为 5 的概率是多少?

- (A) $\frac{61}{216}$ (B) $\frac{367}{1296}$ (C) $\frac{41}{144}$ (D) $\frac{185}{648}$ (E) $\frac{11}{36}$

Problem 21

Regular polygons with 5, 6, 7, and 8 sides are inscribed in the same circle. No two of the polygons share a vertex, and no three of their sides intersect at a common point. At how many points inside the circle do two of their sides intersect?

有 5、6、7 和 8 条边的正多边形都内接在同一个圆中。任意两个多边形没有公共顶点, 并且也没有三条多边形的边相交于一点。由这些多边形的两条边在圆内形成的交点有多少个?

- (A) 52 (B) 56 (C) 60 (D) 64 (E) 68

Problem 22

For each integer $n \geq 2$, let S_n be the sum of all products jk , where j and k are integers and $1 \leq j < k \leq n$. What is the sum of the 10 least values of n such that S_n is divisible by 3?

对于每个整数 $n \geq 2$, 令 S_n 为所有乘积 jk 的总和, 其中 j 和 k 是满足 $1 \leq j < k \leq n$ 的整数。在使得 S_n 能被 3 整除的 n 中, 最小的 10 个数的总和是多少?

- (A) 196 (B) 197 (C) 198 (D) 199 (E) 200

Problem 23

Each of the 5 sides and the 5 diagonals of a regular pentagon are randomly and independently colored red or blue with equal probability. What is the probability that there will be a triangle whose vertices are among the vertices of the pentagon such that all of its sides have the same color?

正五边形的 5 条边和 5 条对角线中的每一条都随机且独立的以相同的概率染为红色或蓝色。存在一个三角形，它的顶点都取自五边形的顶点，并且它的三条边都具有相同颜色的概率是多少？

- (A) $\frac{2}{3}$ (B) $\frac{105}{128}$ (C) $\frac{125}{128}$ (D) $\frac{253}{256}$ (E) 1

Problem 24

A cube is constructed from 4 white unit cubes and 4 blue unit cubes. How many different ways are there to construct the $2 \times 2 \times 2$ cube using these smaller cubes? (Two constructions are considered the same if one can be rotated to match the other.)

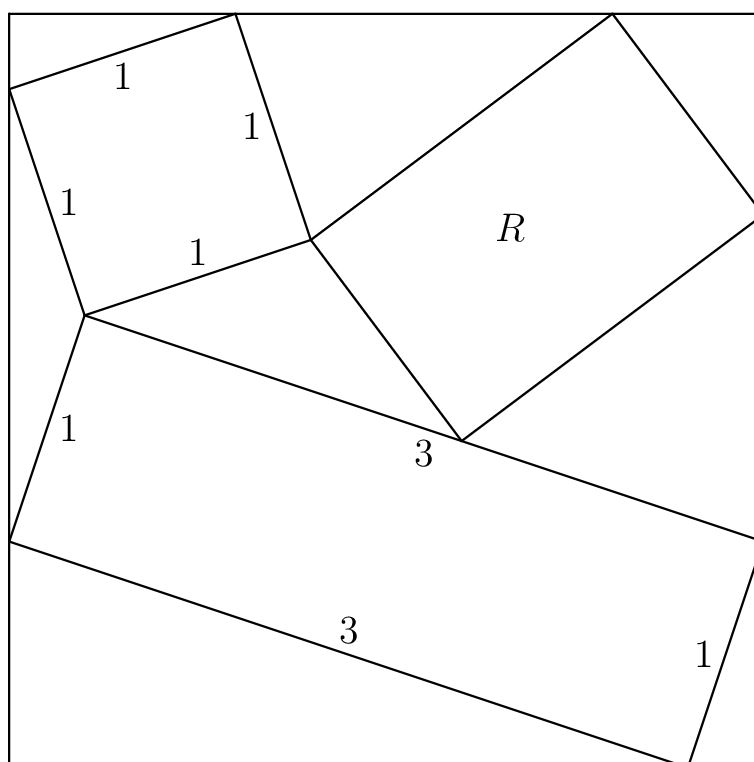
一个立方体由 4 个白色单位立方体和 4 个蓝色单位立方体构成。使用这些较小的立方体来构造该 $2 \times 2 \times 2$ 的立方体，共有多少种不同的方法？（如果一种构型可以通过旋转成为另一种构型，则认为两种构型是相同的。）

- (A) 7 (B) 8 (C) 9 (D) 10 (E) 11

Problem 25

A rectangle with side lengths 1 and 3, a square with side length 1, and a rectangle R are inscribed inside a larger square as shown. The sum of all possible values for the area of R can be written in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. What is $m + n$?

如图所示， 1×3 的矩形、边长为 1 的正方形和矩形 R 内接于一个更大的正方形中。 R 的面积的所有可能值的总和可以写成 $\frac{m}{n}$ 的形式，其中 m 和 n 是互素的正整数。问 $m + n$ 是多少？



- (A) 14 (B) 23 (C) 46 (D) 59 (E) 67

2021Fall AMC 10B Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13
E	B	E	C	E	B	C	C	C	A	B	D	B
14	15	16	17	18	19	20	21	22	23	24	25	
C	D	D	D	E	A	C	E	B	D	A	E	